

Software Testing & Continuous Integration Lab Report

Astrid Jimenez
Group 2 Individual Report

Introduction

This report documents my individual contributions in practicing the fundamentals of software quality assurance through writing unit tests, analyzing code coverage in Python, and automating work with Continuous Integration. The report is appropriately separated into three sections covering each of the three phases, focusing respectively on: Test Coverage, Test-Driven Development, and Continuous Integration.

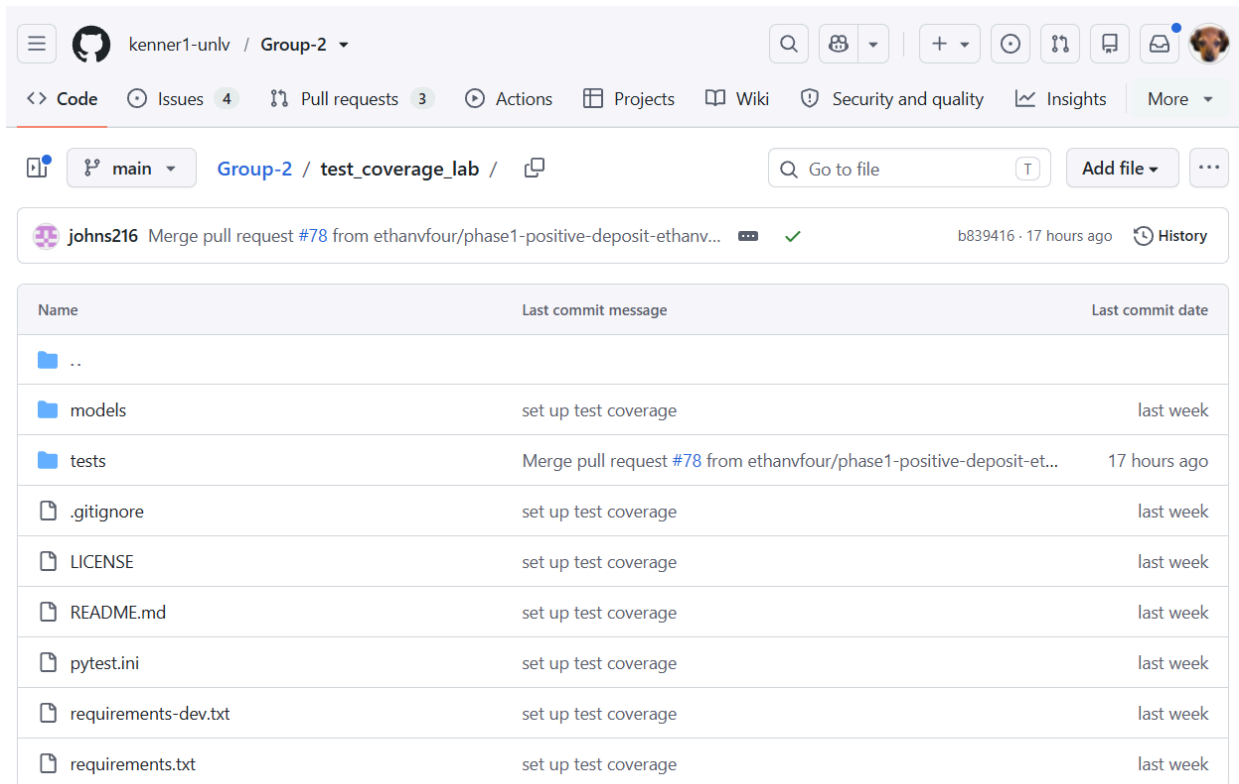
Repository Links and Review Assignments:

- Team repository: [Group 2 GitHub](#)
- Fork repository: [Group 2 Fork](#)
- My assigned reviewer for substantive pull requests: Russell Kennedy (kenner1-unlv)
- My assigned review author: Ethan Guillem (ethanvfour)

Phase 1: Test Coverage

Setup evidence

Before starting my individual contributions, another team member, Michael Podolsky (MikePodo), set up the `test_coverage_lab` folder:



Name	Last commit message	Last commit date
..		
models	set up test coverage	last week
tests	Merge pull request #78 from ethanvfour/phase1-positive-deposit-et...	17 hours ago
.gitignore	set up test coverage	last week
LICENSE	set up test coverage	last week
README.md	set up test coverage	last week
pytest.ini	set up test coverage	last week
requirements-dev.txt	set up test coverage	last week
requirements.txt	set up test coverage	last week

Team Coverage setup Issue link: [Setup Test Coverage Issue 43](#)

Team Coverage PR link: [Setup Test Coverage PR 44](#)

Initial individual work involved creating an appropriate phase1 branch and synchronizing it with the team upstream. I then ensured the test_coverage_lab folder existed in the team repository and installed venv, then activated the virtual environment to run `pip install -r requirements.txt`. This permitted me to run the test coverage suite.

The following screenshot demonstrates a successful run of `pytest --cov=models --cov-report=term-missing`:

```

===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astri/Desktop/cs370/Group-2/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astri/Desktop/cs370/Group-2/test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 3 items

tests/test_account.py::test_account_role_assignment PASSED [ 33%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 66%]
tests/test_account.py::test_password_hashing PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                                Stmts  Miss  Cover   Missing
-----
models/__init__.py                   7      0   100%
models/account.py                   58     24    59%    30, 34, 47-49, 53-56, 60-62, 66-68, 72-76, 94, 98, 102-103
tests/__init__.py                    0      0   100%
tests/test_account.py               43      4    91%    27-30
-----
TOTAL                               108     28    74%

===== 3 passed in 1.24s =====

```

- Link to Phase 1 issue with acceptance criteria in the description: [Phase 1 Issue 64](#)
- Link to PR: [Phase 1 PR 65](#)
- Link to coverage report and results: [Test coverage Report Initial](#)
- Link to re-reviewed coverage report and results: [Test Coverage Report Re-review](#)

Summary of test and behavior verification

Selection of Phase 1 Pool item 7 gave me the responsibility of adding a unit test for 'Account.withdraw()'. The test is to set up an account with a set balance, attempt to withdraw more than the balance holds, to ensure 'withdraw()' raises its 'DataValidationError' flag, and assert at the end of the interaction that the account's balance remains unchanged.

Behavior was verified by running the coverage report suite with `pytest --cov=models --cov-report=term-missing`, demonstrated below, where 'test_withdraw_insuff_funds' was added as an implementation of the test:

```

===== 1 passed in 0.80s =====
(venv) astrid@astrid-VirtualBox:~/Desktop/cs370/Group-2/test_coverage_lab$ pyte
st --cov=models --cov-report=term-missing
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astrid/Des
ktop/cs370/Group-2/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astrid/Desktop/cs370/Group-2/test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 4 items

tests/test_account.py::test_account_role_assignment PASSED [ 25%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 50%]
tests/test_account.py::test_password_hashing PASSED [ 75%]
tests/test_account.py::test_withdraw_insuff_funds PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                               Stmts  Miss  Cover   Missing
-----
models/__init__.py                   7      0   100%
models/account.py                   58     21    64%   30, 34, 47-49, 53-56, 60-62, 66-6
3, 73, 76, 94, 98, 102-103
tests/__init__.py                    0      0   100%
tests/test_account.py               48      4    92%   27-30
-----
TOTAL                               113     25    78%

===== 4 passed in 1.35s =====

```

- Link to technical peer review with comments included: [PR 65 ethanvfour initial review](#)
- Link to re-review technical peer review: [PR 65 ethanvfour re-review](#)

Summary of reviewer feedback and how it was addressed

Ethan's feedback highlighted a missing descriptive docstring under my test function's definition, a typo of `test\_account.py`, and merge conflicts. The first changes I implemented were an addition of a descriptive docstring under "test_withdraw_insuff_funds()" and fixing the previous typo "test_accounts.py" under the PR description to "test_account.py". Once those were completed, it was simple to run `Git merge main` and ensure the merge was clean. No conflicts were reported, and a clean merge was evident after running the command as shown:

```

git merge main
Auto-merging test_coverage_lab/tests/test_account.py
Merge made by the 'ort' strategy.
 .github/workflows/ci.yml | 13 ++++--
 README.md | 22 ++++++++
 ci_lab/tests/test_counter.py | 5 +-
 tdd_lab/src/counter.py | 21 ++++++++
 tdd_lab/tests/test_counter.py | 39 ++++++++
 test_coverage_lab/tests/test_account.py | 83
 ++++++
 6 files changed, 177 insertions(+), 6 deletions(-)

```

After merging, I ran pytest where results provided evidence that the suite still ran as expected:

```

t --cov=models --cov-report=term-missing
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astrid/Desktop/cs370/Group-2/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astrid/Desktop/cs370/Group-2/test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 8 items

tests/test_account.py::test_account_role_assignment PASSED [ 12%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 25%]
tests/test_account.py::test_to_dict PASSED [ 37%]
tests/test_account.py::test_required_fields PASSED [ 50%]
tests/test_account.py::test_password_hashing PASSED [ 62%]
tests/test_account.py::test_account_deactivation_and_reactivation PASSED [ 75%]
tests/test_account.py::test_valid_withdrawal PASSED [ 87%]
tests/test_account.py::test_withdraw_insuff_funds PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                               Stmts  Miss  Cover   Missing
-----
models/__init__.py                   7      0   100%
models/account.py                   58     13    78%   30, 47-49, 60-62, 66-68, 73, 102-103
tests/__init__.py                     0      0   100%
tests/test_account.py               74      0   100%
-----
TOTAL                               139     13    91%
===== 8 passed in 1.21s =====

```

Feedback and final status

Ethan approved the Phase 1 pull request after confirming the merge conflicts were resolved, a descriptive docstring was added under the test function, and the typo was corrected within the PR description. No further changes were requested, and the pull request was approved and merged.

The committed and approved PR is linked: [Add test for withdrawal with insufficient funds #65](#)

Technical peer review

I reviewed Russell Kennedy's Phase 1 pull request who covered Phase 1 pool item 9 and focused on adding tests for account deactivation and reactivation behavior in `test_account.py`.

- Link to PR: [kenner1-unlv PR 61](#)
- Link to my review: [technical review of PR 61](#)

The pull request was approved under the conditions that:

- The tests exercised the intended behavior of item 9,
- Assertions were specific and meaningful,
- Running the coverage test with the new function `test_account_deactivation_and_reactivation()` was successful and proved to exercise the methods needed to be tested properly,
- No regressions or merge conflicts were present at the time of review.

Phase 2: Test-Driven Development (TDD)

Setup Evidence

Setup of the Test-Driven Development environment was initialized by Glen Testamark (geeklay). The `tdd_lab` folder is evident below:

☰

kenner1-unlv / Group-2

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<> Code

🕒 Issues 4

🔗 Pull requests 3

👤 Actions

📁 Projects

📖 Wiki

🛡️ Security and quality

📈 Insights

More

📁

main

Group-2 / tdd_lab /

📄

🔍 Go to file

T

Add file

⋮

👤 Peekans

Used a valid name for test ✓

34e0b11 · 1 hour ago

🕒 History

Name	Last commit message	Last commit date
..		
doc	initial tdd_lab commit	last week
src	Add counter_exists check to return 404 (GREEN)	2 hours ago
tests	Used a valid name for test	1 hour ago
.gitignore	initial tdd_lab commit	last week
LICENSE	initial tdd_lab commit	last week
README.md	initial tdd_lab commit	last week
pytest.ini	initial tdd_lab commit	last week
requirements.txt	initial tdd_lab commit	last week

My individual work began with synchronizing a new branch specified for phase2 work with the upstream repository, installing the `requirements.txt` for `tdd_lab`, and running ``pytest—cov=src``, where a successful run of the suite was provided in my work environment terminal:

```
(venv) astrid@astrid-VirtualBox:~/Desktop/cs370/Group-2/tdd_lab$ pytest --cov=src
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astrid/Desktop/cs370/Group-2/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astrid/Desktop/cs370/Group-2/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 2 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py         3      0   100%
src/counter.py        15      1    93%    19
src/status.py          6      0   100%
-----
TOTAL                  24      1    96%

===== 2 passed in 0.33s =====
```

- Link to my issue and acceptance criteria: [TDD Issue 66](#)
- Link to my TDD PR with test case included in the description: [TDD PR 67](#)

Brief explanation of feature being tested

Claiming Phase 2 pool item 4 came with the responsibility to implement the increment feature of the Counter API utilizing the TDD cycle. The expected behavior of `PUT /counters/<name>` should increment an existing counter by 1 and return the proper updated value with `HTTP\_200\_OK`.

Evidence of RED/GREEN/REFACTOR and test results

Link to the evidence and test results: [TDD PR 67 Description](#)

Before implementing the feature, a written test for pool item 4 was created for the expected behavior, provoking the RED phase. `test\_increment\_counter` was added to `tests/test\_counter.py`, which creates a counter, increments it with PUT, and asserts, by checking against a 200 value, that the counter was increased to 1.

The added code is shown:

```

46     def test_increment_counter(self, client):
47         """It should increment an existing counter's value"""
48         # Starting point: creating a counter starting at 0
49         client.post('/counters/inc')
50
51         # Implementation via PUT
52         result = client.put('/counters/inc')
53
54         # Assert: request success and value increased
55         assert result.status_code == status.HTTP_200_OK
56         data = result.get_json()
57         assert data['inc'] == 1
58

```

With no PUT route present, Flask returned 405 Method Not Allowed, demonstrating the test failed as expected:

```

===== FAILURES =====
_____ TestCounterEndpoints.test_increment_counter _____

self = <tests.test_counter.TestCounterEndpoints object at 0x73169e697bf0>
client = <FlaskClient <Flask 'src.counter'>>

    def test_increment_counter(self, client):
        """It should increment an existing counter's value"""
        # Starting point: creating a counter starting at 0
        client.post('/counters/inc')

        # Implementation via PUT
        result = client.put('/counters/inc')

        # Assert: request success and value increased
>       assert result.status_code == status.HTTP_200_OK
E       assert 405 == 200
E       + where 405 = <WrapperTestResponse streamed [405 METHOD NOT ALLOWED]>
E       +   .status_code
E       + and 200 = status.HTTP_200_OK

tests/test_counter.py:55: AssertionError
===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py        3      0   100%
src/counter.py       15      1    93%   19
src/status.py         6      0   100%
TOTAL                 24      1    96%

===== short test summary info =====
FAILED tests/test_counter.py::TestCounterEndpoints::test_increment_counter - as
sert 405 == 200
===== 1 failed, 2 passed in 0.42s =====

```

Minimal code was added to `src/counter.py::increment\_counter` by adding a PUT route to increment the counter and return 200.

The added code is shown:

```
28 @app.route('/counters/<name>', methods=['PUT'])
29 def increment_counter(name):
30     """Increment an existing counter"""
31     COUNTERS[name] += 1
32     return jsonify({name: COUNTERS[name]}), status.HTTP_200_OK
33
```

When the suite was re-run, the GREEN phase was initialized with all tests passing:

```
(venv) astrid@astrid-VirtualBox:~/Desktop/cs370/Group-2/tdd_lab$ pytest --cov=src
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astrid/Desktop/cs370/Group-2/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astrid/Desktop/cs370/Group-2/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 3 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 33%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [ 66%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py         3      0   100%
src/counter.py        19      1    95%    19
src/status.py          6      0   100%
-----
TOTAL                  28      1    96%

===== 3 passed in 0.40s =====
```

REFACTOR was not necessary, as the previous code was short and clear-cut; extracting or restructuring it would have added complexity without any real benefit.

Summary of reviewer feedback and how it was addressed

- Link to Ethan's initial review: [Review PR 67](#)
- Link to Ethan's re-review: [Re-Review PR 67](#)

Initial feedback reported verification of the tests, implementation, and justifications. Ethan brought to my attention that for future TDD work, RED/GREEN/REFACTOR phases could be committed to the same PR as separate commits to provide a test-first sequence independently auditable in the Git history. Any changes were related to merge conflicts.

- Link to my response to the initial review: [Author response PR 67](#)

In response to the feedback, I familiarized myself with the process of implementing separate commits for each TDD phase to ensure future work would not be met with the same issue. Regarding merge conflicts, I merged my fork into the main branch, where the following conflicts were highlighted:

```
Auto-merging tdd_lab/src/counter.py
CONFLICT (content): Merge conflict in tdd_lab/src/counter.py
Auto-merging tdd_lab/tests/test_counter.py
CONFLICT (content): Merge conflict in
tdd_lab/tests/test_counter.py
Automatic merge failed; fix conflicts and then commit the
result.
```

In response to the conflicts, I reviewed the conflict markers Git inserted in `counter.py` and `test\_counter.py`, confirmed both additions were separate, and resolved the conflict by keeping both. After resolution, I reran the suite, where a successful run was evident:

```

platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astrid/Desktop/cs370/Group-2/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astrid/Desktop/cs370/Group-2/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 6 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 16%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 33%]
tests/test_counter.py::TestCounterEndpoints::test_list_counters PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 66%]
]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 83%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [100%]

===== tests coverage =====
----- coverage: platform linux, python 3.12.3-final-0 -----
Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py       3        0   100%
src/counter.py       32        1    97%    24
src/status.py         6        0   100%
-----
TOTAL                 41        1    98%
===== 6 passed in 0.43s =====

```

Feedback and final status

Ethan approved the Phase 2 pull request after re-reviewing, ensuring the latest head followed the merge from main, differences focused on the previous merge conflicts, and my implementation was well within the scope of my responsibilities without adding any conflicts. Once approved, the pull request was merged into the main repository.

The committed and approved PR is linked: [Add increment counter feature with TDD #67](#)

Technical peer review

I reviewed Russell's Phase 2 pull request where he worked on pool item 8, adding endpoint resets to all counters. His test implementation verified reset behavior with counters starting at different nonzero values, then checked that they were zero after. RED/GREEN/REFACTOR phases were all committed separately, providing a history of implementation within Git's history. A genuine refactor was established through creation of `reset\_counters()`, which separates HTTP handling from the reset (endpoint) logic. Preserving the behavior of the GREEN phase and a

successful run to the suite. My review included emphasis on a merge conflict in `test\_counter.py` and requested that Russell sync with upstream and resolve it, while providing evidence the suite still passes after resolution.

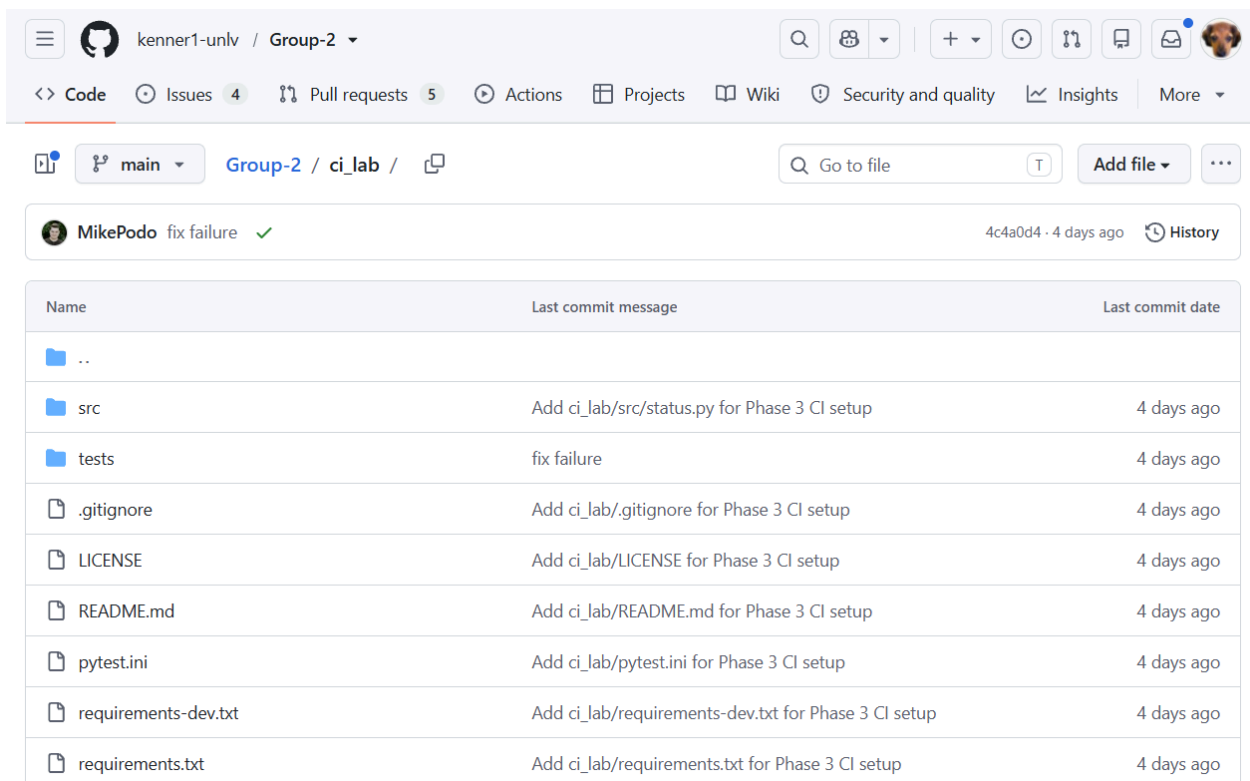
In my re-review, I ensured a clean merge was conducted with a fresh run of the suite provided. Verifying that the files altered retained the merge-conflicting code and the PR's reset test, endpoint, and `reset\_counters()` function. After this re-review, the pull request was approved and merged into main.

- Link to Russell's Merged Pull Request: [Add endpoint to reset all counters #62](#)
- Link to my Initial review: [PR 62 Review](#)
- Link to my re-review: [PR 62 Re-Review](#)

Phase 3: Continuous Integration

Setup evidence

Phase 3's starter files and setup were completed by Russell Kennedy (kenner1-unlv); the folder `ci\_lab` is demonstrated below:



Name	Last commit message	Last commit date
..		
src	Add ci_lab/src/status.py for Phase 3 CI setup	4 days ago
tests	fix failure	4 days ago
.gitignore	Add ci_lab/.gitignore for Phase 3 CI setup	4 days ago
LICENSE	Add ci_lab/LICENSE for Phase 3 CI setup	4 days ago
README.md	Add ci_lab/README.md for Phase 3 CI setup	4 days ago
pytest.ini	Add ci_lab/pytest.ini for Phase 3 CI setup	4 days ago
requirements-dev.txt	Add ci_lab/requirements-dev.txt for Phase 3 CI setup	4 days ago
requirements.txt	Add ci_lab/requirements.txt for Phase 3 CI setup	4 days ago

- Link to CI_lab Setup Issue: [Setup Phase 3 #51](#)
- Link to CI_lab Setup PR: [Setup PR #52](#)

To set up my environment, I created an appropriate phase3 branch, switched to it, and synced it with the main upstream. Then I installed the required `requirements.txt` file and verified the setup by running `pytest --cov=src --cov-report=term-missing`, ensuring all 22 starter tests pass at roughly 95% coverage.

First run of `pytest --cov=src --cov-report=term-missing` after sync:

```
(venv) astrid@astrid-VirtualBox:~/Desktop/cs370/Group-2/ci_lab$ pytest --cov=src
--cov-report=term-missing
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/astrid/Desktop/cs370/Group-2/ci_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/astrid/Desktop/cs370/Group-2/ci_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 4%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 9%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 13%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 18%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 22%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_updating_non_existent_counter PASSED [ 27%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 31%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_deleting_non_existent_counter PASSED [ 36%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 40%]
tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 54%]
```

```

tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 59%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 63%]
]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [
68%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_negative_counter_value
s PASSED [ 72%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [
77%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_resetting_non_existent
_counter PASSED [ 81%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters P
ASSED [ 86%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshol
d PASSED [ 90%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold P
ASSED [ 95%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [
100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py       3      0   100%
src/counter.py       87      5    94%    71, 85, 96, 99-100
src/status.py         7      0   100%
-----
TOTAL                 97      5    95%
===== 22 passed in 0.68s =====
(venv) astrid@astrid-VirtualBox:~/Desktop/cs370/Group-2/ci_lab$

```

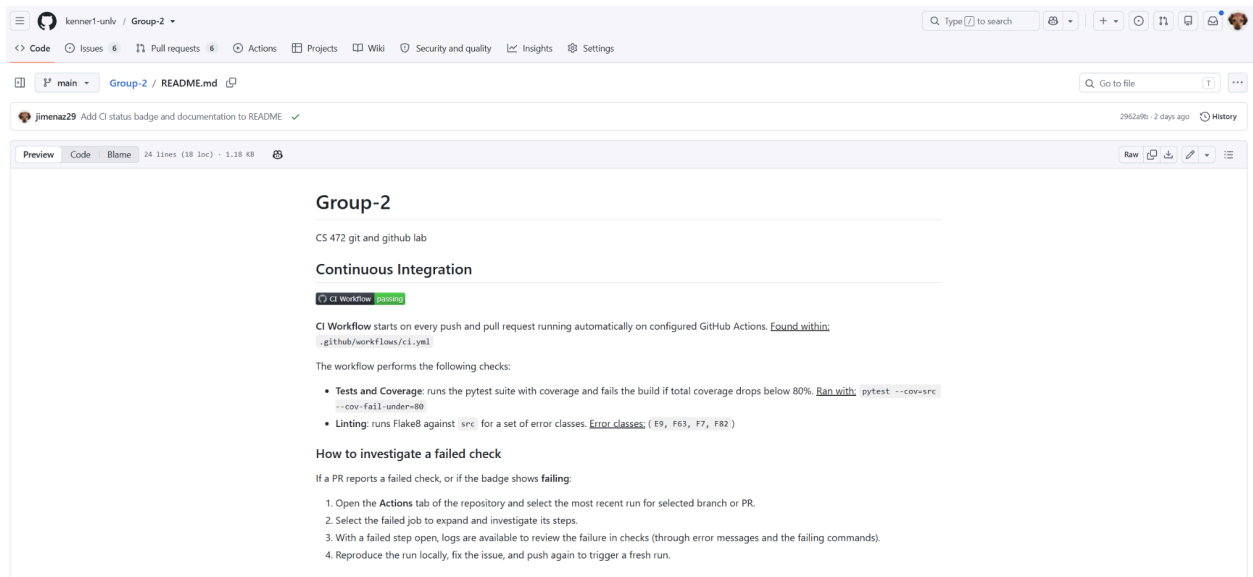
CI Status Badge and Documentation to README

My phase 3 responsibilities fell under enhancement 6: adding a CI status badge, CI documentation, and an explanation of how contributors can investigate a failed check, added to the repository's root

'README.md'.

- Link to my issue: [CI visibility: status badge and documentation #86](#)
- Link to PR: [Add CI status badge and documentation to README](#)
- Link to root repository's 'README.md': [Group-2 README](#)

Evidence of CI Status Badge and Documentation in README:



First successful run:

Workflow: Add CI status badge and documentation to README #62

Summary

All jobs

- build (3.9)
- build (3.10)
- build (3.11)

Run details

Usage

Workflow file

Triggered via pull request 2 days ago

Success Total duration: **26s** Artifacts: **-**

ci.yml on: pull_request

Matrix build

3 jobs completed

Show all jobs

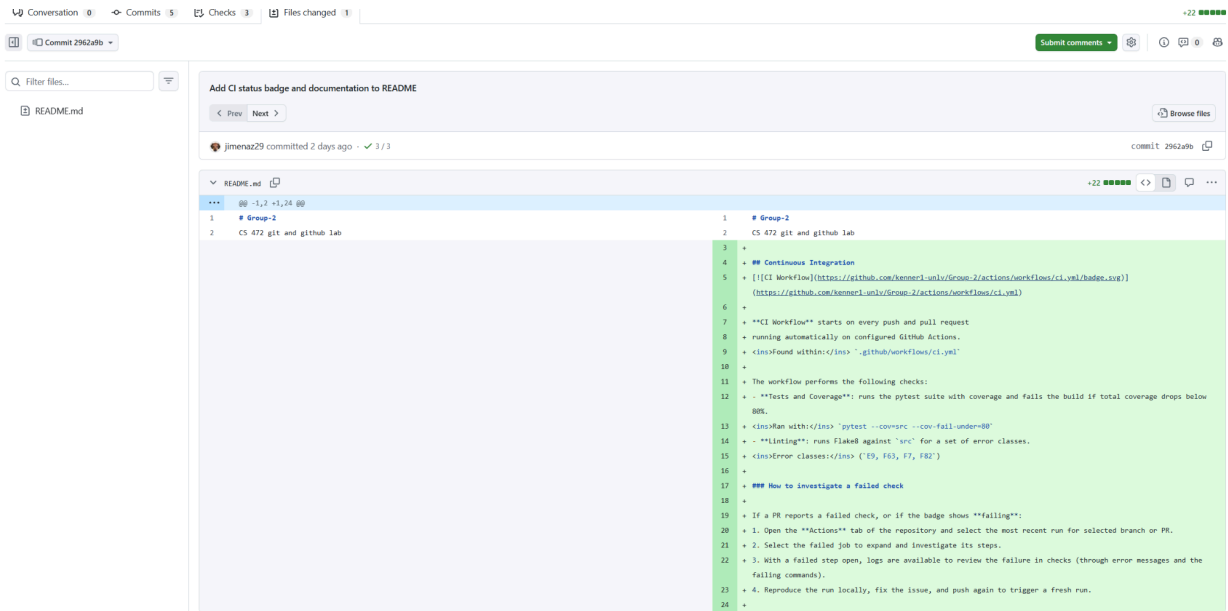
Annotations

3 warnings

- build (3.11)**
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-n->
[Show more](#)
- build (3.10)**
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-n->
[Show more](#)
- build (3.9)**
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-n->
[Show more](#)

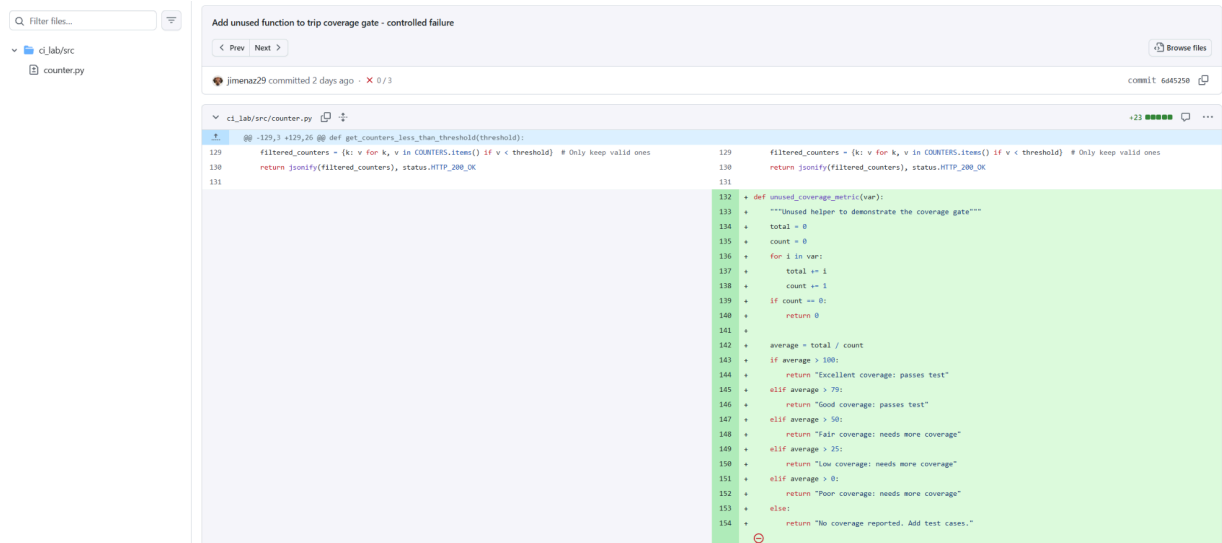
- Link to the first commit only changing the 'README' file: [PR 88 Add CI status badge and documentation to 'README'](#)

Documentation of differences:



After adding the CI status badge and documentation to the repository's root 'README'. I created 'unused_coverage_metric(var)' in 'ci_lab/src/counter.py', which would trigger a controlled failure, as the coverage rate with the new function would drop below the 80% threshold documented in the 'README' file. This controlled failure exemplified verification of the README documentation and demonstrated familiarity with the CI workflow process.

The function definition of 'unused_coverage_metric(var)' is shown below:



- Link to the controlled-failure CI run: [PR 88 controlled failure](#)

Red run phase:

Add CI status badge and documentation to README #63

Re-run jobs

Summary

All jobs

build (3.9)

build (3.10)

build (3.11)

Run details

Usage

Workflow file

Triggered via pull request 2 days ago

Status

Total duration

Artifacts

jimenaz29 synchronize #65

3jimenaz29:phase3-ci-vizibill

Failure

27s

-

ci.yml

on: pull_request

Metric: build

3 jobs completed

Show all jobs

Annotations

3 errors and 3 warnings

build (3.10)

Process completed with exit code 1.

build (3.11)

Process completed with exit code 1.

build (3.9)

Process completed with exit code 1.

build (3.10)

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-n->

Show more

build (3.11)

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-n->

Show more

build (3.9)

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-19-deprecation-of-n->

Show more

Investigation of failure points to the coverage rate not meeting the 80% threshold:

```

17 collecting ... collected 22 items
18
19 tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 4%]
20 tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 9%]
21 tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 13%]
22 tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 18%]
23 tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 22%]
24 tests/test_counter.py::TestCounterEndpoints::test_prevent_updating_non_existent_counter PASSED [ 27%]
25 tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 31%]
26 tests/test_counter.py::TestCounterEndpoints::test_prevent_deleting_non_existent_counter PASSED [ 36%]
27 tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 40%]
28 tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 45%]
29 tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 50%]
30 tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 54%]
31 tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 59%]
32 tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 63%]
33 tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 68%]
34 tests/test_counter.py::TestCounterEndpoints::test_prevent_negative_counter_values PASSED [ 72%]
35 tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 77%]
36 tests/test_counter.py::TestCounterEndpoints::test_prevent_resetting_non_existent_counter PASSED [ 81%]
37 tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 86%]
38 tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshold PASSED [ 90%]
39 tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold PASSED [ 95%]
40 tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [100%]
41 ERROR: Coverage failure: total of 79 is less than fail-under=80
42
43
44 ===== tests coverage =====
45 _____ coverage: platform linux, python 3.10.21-final-0 _____
46
47 Name                Stmts  Miss  Cover   Missing
48 -----
49 src/__init__.py       3      0   100%
50 src/counter.py       107     24    78%   71, 85, 96, 99-100, 134-154
51 src/status.py         7      0   100%
52 -----
53 TOTAL                 117     24    79%
54 FAIL Required test coverage of 80% not reached. Total coverage: 79.49%
55 ===== 22 passed in 0.33s =====
56 Error: Process completed with exit code 1.
```

To return to a green running state, the function was omitted from the files, leaving the altered `README.md` file as the only file committed in the pull request.

- Link to fix commit: [PR 88 remove 'unused coverage metric\(\)'](#)

Snapshot of PR difference after removing new function:

```
Removed unused function to restore coverage above gate  
jimenaz29 committed 2 days ago · ✓ 3/3  
commit 5a77606  
c:\lab\src\counter.py  
@@ -128,27 +128,4 @@ def get_counters_less_than_threshold(threshold):  
128     """Get all counters with values less than the given threshold"""  
129     filtered_counters = {k: v for k, v in COUNTERS.items() if v < threshold} # Only keep valid ones  
130     return jsonify(filtered_counters), status.HTTP_200_OK  
131  
132 - def unused_coverage_metric(var):  
133 -     """Unused helper to demonstrate the coverage gate"""  
134 -     total = 0  
135 -     count = 0  
136 -     for i in var:  
137 -         total += 1  
138 -         count += 1  
139 -     if count == 0:  
140 -         return 0  
141 -  
142 -     average = total / count  
143 -     if average > 100:  
144 -         return "Excellent coverage: passes test"  
145 -     elif average > 75:  
146 -         return "Good coverage: passes test"  
147 -     elif average > 50:  
148 -         return "Fair coverage: needs more coverage"  
149 -     elif average > 25:  
150 -         return "Low coverage: needs more coverage"  
151 -     elif average > 0:  
152 -         return "Poor coverage: needs more coverage"  
153 -     else:  
154 -         return "No coverage reported. Add test cases."  
128     """Get all counters with values less than the given threshold"""  
129     filtered_counters = {k: v for k, v in COUNTERS.items() if v < threshold} # Only keep valid ones  
130     return jsonify(filtered_counters), status.HTTP_200_OK
```

Re-run of the complete suite was now successful:

Workflow: Merge pull request #67 from jimenaz29/phase2-increment-counter #79
Triggered via push 18 hours ago
Status: Success
Total duration: 47s
Artifacts: 1
Re-run all jobs

Summary
All jobs

- Lint
- Test (Python 3.9)
- Test (Python 3.10)
- Test (Python 3.11)
- Coverage

Run details
Usage
Workflow file

ci.yml
on: push

- Lint 18s
- Coverage 20s

Matrix: test

- 3 jobs completed

Annotations
5 warnings and 5 notices

Test (Python 3.10)
Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/cache@v4, actions/checkout@v4, actions/setup-python@v5. For more information see: <https://github.blog/changelog/2025-09-1->

Summary of reviewer feedback and how it was addressed

Ethan verified that the pull request accurately described and documented the completion of Enhancement 6, including Git history and an implementation of the controlled-failure run. No conflict issues were present, and the pull request was approved and merged upon the first review.

- Link to Ethan's review: [PR 88 Review](#)

Technical peer review

Reviewing Russell's Phase 3 work fell under the responsibilities of Enhancement 7: Split CI workflow into lint, test, and coverage jobs. My technical review verified separate Lint, Test, and Coverage jobs, each set up its environment and ran its respective checks. The controlled failure involved minimal adjustments where Flake8 pointed to `nonexistent\_directory`, failing only the Lint job, verified by the separate commit. When changes were reverted, a green run was evident for all three splits. The only blocker to approval was a merge conflict, which was resolved; the suite was then re-run and verified to remain green. Once changes were made and verified, the pull request was approved and merged.

- Link to Russell's committed PR: [Split CI workflow into lint, test, and coverage jobs #57](#)
- Link to my initial review: [PR #57 Review](#)
- Link to first re-review: [PR #57 Re-Review](#)
- Link to second re-review: [PR #57 Final Review](#)

Reflection

This GitHub lab helped me understand the specifications of continuous integration, automated testing, coverage and quality gates, CI failure diagnosis, technical peer review, collaborative software development, and the importance of documentation. The greatest emphasis revolved around team coordination and communication; this lab helped strengthen those qualities within my team. I learned how separate commits can provide further documentation, verification, and integrity to individual pull requests, a very handy tool to utilize in the future. Overall, this lab served as an environment for trial, error, and learning to understand workflow and coordination better than I had known in the beginning.